

# DHA SUFFA UNIVERSITY

## Department of Computer Science



### CryptoBuddy

### Cryptocurrency Prediction System

### Final Year Project Report

Version 1.0

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## Certificate of Approval

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No part of this report has been submitted anywhere else for any other degree.

This report is submitted to the Department of Computer Science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science

at

DHA Suffa University (DSU)

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## Authors' Declaration

We declare that this project report was carried out in accordance with the rules and regulations of the DHA Suffa University (DSU). The work is original except where indicated by special references in the text and no part of the report has been submitted for any other degree. The report has not been presented to any other University for examination.

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We, < Dev Yogesh>, <Zona khan>, < Mirza Omer Baig > and <Hattaf Ali>, solemnly declare that the work presented in the Final Year Project Report titled <Project title> has been carried out solely by ourselves with no significant help from any other person except few of those which are duly acknowledged. I confirm that no portion of our report has been plagiarized and any material used in the report from other sources is properly referenced.

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# CHAPTER 1

## INTRODUCTION

### 1.1 Description about Project

CryptoBuddy is a real-time cryptocurrency trading signal system developed as a Final Year Project at DHA Suffa University. The system leverages machine learning, technical analysis, and sentiment intelligence to generate actionable trading signals — BUY, SELL, or HOLD — for three of the world's most traded cryptocurrencies: Bitcoin (BTCUSDT), Ethereum (ETHUSDT), and Solana (SOLUSDT).

The core objective of CryptoBuddy is to provide retail cryptocurrency traders with data-driven, explainable signals that go beyond simple rule-based alerts. Unlike conventional trading bots that act on a single indicator crossing a threshold, CryptoBuddy employs a multi-layered decision pipeline: it first trains a supervised machine learning classifier on 61 engineered technical features derived from historical hourly price data, then validates any predicted signal against six independent technical confirmation checks, and finally overlays real-time market sentiment before outputting a final decision.

The system pipeline is structured as follows: live Open-High-Low-Close-Volume (OHLCV) data is fetched from the Binance exchange API at one-hour intervals. From this data, 61 technical indicators are computed spanning momentum, trend, volatility, volume, candle pattern, and multi-horizon return domains. A GradientBoosting classifier — trained independently for each coin — then predicts whether the next hourly candle is likely to move up (BUY), down (SELL), or sideways (HOLD). This model prediction is then cross-validated by the multi-signal confirmation engine, which checks six independent technical conditions. Finally, real-time sentiment from the Crypto Fear & Greed Index and news RSS feeds, alongside a risk event detector, is applied before the signal is displayed on the React.js frontend dashboard.

The frontend presents each coin's final signal, confidence level, Fear & Greed score, news sentiment, risk score, confirmation score, and detailed breakdown of probabilities and confirmation reasons in a clean single-page interface, giving traders full transparency into why each signal was generated.

### 1.2 Details about the Domain



Cryptocurrency markets represent one of the most unique and challenging domains for financial prediction. Unlike traditional stock markets, crypto markets operate 24 hours a day, 7 days a week, 365 days a year with no circuit breakers, no trading halts, and no market maker obligations. This continuous operation means price movements can compound rapidly at any time of day or night without institutional intervention.

The three cryptocurrencies targeted by CryptoBuddy — Bitcoin, Ethereum, and Solana — collectively represent a significant portion of the total cryptocurrency market capitalisation. Bitcoin is the oldest and most liquid, widely regarded as digital gold and a macro asset. Ethereum is the leading smart contract platform and underpins the majority of decentralised applications (dApps) and DeFi protocols. Solana is a high-throughput blockchain known for its speed and low transaction costs, making it particularly volatile relative to its peers.

Price prediction in cryptocurrency markets is uniquely challenging for several reasons. First, crypto markets are heavily influenced by retail sentiment, social media narratives, and influencer commentary in ways that traditional markets are not. A single tweet from a prominent figure can move Bitcoin's price by several percent within minutes. Second, the market is global and fragmented across hundreds of exchanges, meaning price discovery is inherently noisy. Third, the asset class is relatively young, meaning historical patterns are less established than in equity markets.

At a one-hour prediction horizon, the task is particularly difficult. The Efficient Market Hypothesis (EMH) suggests that at short horizons, price movements are close to random as all publicly available information is rapidly priced in. However, order flow imbalances, momentum effects, and volatility clustering do create exploitable short-term patterns, which is what technical indicator-based machine learning systems attempt to capture.

Trading signals in this context are classified into three categories: BUY (enter or hold a long position), SELL (exit or short), and HOLD (take no action). The HOLD class is critically important — it represents the model's acknowledgment of uncertainty, which is arguably more valuable than a wrong BUY or SELL signal in live trading.

## 1.3 Relevant Background

Technical analysis is the practice of using historical price and volume data to forecast future price movements. Indicators derived from this data — such as the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, and Average True Range (ATR) —

serve as the primary features in CryptoBuddy's machine learning pipeline. These indicators are widely used by traders and have been extensively studied in academic literature as inputs to predictive models.

Sentiment analysis in financial markets refers to the extraction of market mood from textual data sources such as news articles, social media, and community forums. In cryptocurrency markets, sentiment plays an outsized role compared to traditional finance, as the asset class lacks fundamental valuation anchors (like P/E ratios for stocks). The Crypto Fear & Greed Index — used in CryptoBuddy as a real-time sentiment signal — aggregates multiple sentiment indicators into a single score from 0 (Extreme Fear) to 100 (Extreme Greed), providing a market-wide mood reading that updates every hour.

Gradient Boosting is an ensemble machine learning technique that builds a sequence of decision trees, where each tree corrects the errors of the previous one. It is particularly well-suited to tabular financial data because it handles non-linear feature interactions naturally, is robust to outliers, and does not require feature scaling. Unlike deep learning approaches, it is computationally feasible on consumer hardware and produces interpretable feature importance scores.

Multi-signal confirmation is a risk management principle borrowed from professional trading: no single indicator or model should be trusted in isolation. CryptoBuddy implements this by requiring that a minimum number of independent technical checks agree with the model's prediction before a BUY or SELL signal is issued. Signals that fail to achieve sufficient confirmation are downgraded to HOLD, reducing false positives at the cost of signal frequency — a deliberate precision-over-recall design choice.

## **Chapter 2: Relevant Background & Definitions**

### **2.1 Cryptocurrency and Blockchain**

A cryptocurrency is a digital or virtual currency that uses cryptographic techniques to secure transactions and control the creation of new units. Unlike traditional fiat currencies, cryptocurrencies operate on decentralised networks — typically blockchains — that are maintained by a distributed network of nodes rather than a central authority. Bitcoin, introduced in 2009, was the first cryptocurrency and remains the largest by market capitalisation. Ethereum introduced programmable smart contracts in 2015, enabling decentralised applications. Solana, launched in 2020, offers high throughput (up to 65,000 transactions per second) and is known for its speed and low fees.

### **2.2 OHLCV Data**

OHLCV stands for Open, High, Low, Close, Volume — the five core data points recorded for each time interval (candle) in financial markets. The Open is the price at which the asset traded at the start of the interval; the High and Low are the maximum and minimum prices reached during the interval; the Close is the final price; and Volume is the total quantity of the asset traded. CryptoBuddy uses one-hour OHLCV candles fetched from the Binance API, creating a time series with one row per hour per coin.

### **2.3 Technical Indicators**

Technical indicators are mathematical transformations of OHLCV data used to identify patterns and trends. The following indicators are used as features in CryptoBuddy:

**Relative Strength Index (RSI):** A momentum oscillator measuring the speed and magnitude of price changes, producing a value between 0 and 100. Values above 70 indicate overbought conditions; below 30 indicate oversold. CryptoBuddy computes RSI at 7, 14, and 21-period windows.

**MACD (Moving Average Convergence Divergence):** The difference between a 12-period and 26-period exponential moving average (EMA). The MACD histogram (MACD minus signal line) is particularly useful as a momentum confirmation signal.

**Bollinger Bands:** A volatility indicator consisting of a 20-period moving average with upper and lower bands at 2 standard deviations. The band position (where price sits relative to the bands) and band width (volatility measure) are used as features.

**Average True Range (ATR):** A volatility indicator measuring the average range of price movement over 14 periods. ATR expressed as a percentage of price (atr\_pct) was found to be the most important feature in CryptoBuddy's models, with 50.9% importance for BTCUSDT.

**Stochastic Oscillator:** Compares the closing price to its price range over 14 periods, producing %K and %D lines. Useful for identifying momentum reversals.

**Exponential Moving Average (EMA):** A weighted moving average giving more weight to recent prices. CryptoBuddy uses 20-period and 50-period EMAs, their crossover, and price position relative to each EMA.

## **2.4 Supervised Machine Learning Classification**

Supervised machine learning involves training a model on labelled examples to learn a mapping from input features to output classes. In CryptoBuddy, the model is trained on historical candles where the label (BUY/SELL/HOLD) has been determined by the actual future price movement. Classification is the task of assigning an input to one of a fixed set of discrete categories — in this case, three: SELL (0), BUY (1), HOLD (2).

## **2.5 GradientBoostingClassifier**

Gradient Boosting is an ensemble technique that builds an additive model of decision trees sequentially. Each new tree is trained to correct the residual errors of the existing ensemble. The learning\_rate parameter controls the contribution of each tree; lower values (e.g. 0.05) require more trees but generalise better. max\_depth controls tree complexity; shallow trees (3-5 levels) prevent overfitting on noisy financial data. GradientBoosting outperforms Random Forests on many tabular datasets because it explicitly optimises the loss function rather than averaging uncorrelated trees.

## **2.6 TimeSeriesSplit Cross-Validation**

Standard k-fold cross-validation randomly partitions data into folds, which is inappropriate for time series because it allows future data to inform past predictions (data leakage). TimeSeriesSplit

preserves temporal order by using only past data for training and future data for validation in each fold. For example, with 3 splits on a 15,000-row dataset: Fold 1 trains on rows 1-5000 and validates on 5001-10000; Fold 2 trains on 1-10000 and validates on 10001-15000. This ensures evaluation reflects real-world deployment conditions.

## 2.7 Sentiment Analysis

Sentiment analysis is the computational process of identifying and quantifying the emotional tone of textual data. CryptoBuddy employs two sentiment sources: VADER (Valence Aware Dictionary and sEntiment Reasoner), a rule-based sentiment analyser calibrated for social media text, is applied to cryptocurrency news headlines from RSS feeds. The Crypto Fear & Greed Index is a composite sentiment indicator that aggregates volatility, market momentum, social media activity, and surveys into a 0-100 score, updated hourly.

## 2.8 Class Imbalance

Class imbalance occurs when training data contains significantly more examples of some classes than others. In financial classification, if HOLD represents 60% of samples, a naive model can achieve 60% accuracy by always predicting HOLD — useless in practice. CryptoBuddy addresses this using `compute_sample_weight('balanced')`, which assigns higher training weights to underrepresented classes, and uses Macro F1 (which weights all classes equally) as the optimisation metric rather than accuracy.

## 2.9 Precision vs Recall Tradeoff

Precision measures what fraction of predicted positives are actually positive (signal reliability). Recall measures what fraction of actual positives were detected (signal coverage). In trading, false signals are more harmful than missed signals: acting on a wrong BUY signal causes a loss, while missing a BUY signal is merely a missed opportunity. CryptoBuddy therefore prioritises precision through multi-signal confirmation — requiring agreement from multiple independent checks before issuing a signal — accepting lower recall in exchange for higher confidence in signals that are issued.

## **Chapter 3: Literature Review & Related Work**

### **3.1 Literature Review**

#### **3.1.1 Machine Learning for Cryptocurrency Prediction**

The application of machine learning to cryptocurrency price prediction has grown substantially since Bitcoin's mainstream adoption. Nakano et al. (2018) demonstrated that artificial neural networks trained on high-frequency Bitcoin price data could achieve modest but statistically significant predictive accuracy above random baselines, establishing the feasibility of ML-based crypto prediction. Their work highlighted the importance of using appropriate evaluation metrics beyond accuracy for imbalanced financial datasets.

Ji et al. (2019) conducted a comprehensive study comparing multiple ML algorithms — including Random Forest, Support Vector Machine, and Gradient Boosting — for predicting the directional movement of six major cryptocurrencies. Their findings showed that ensemble methods, particularly Gradient Boosting, consistently outperformed single classifiers on tabular OHLCV feature sets, which directly informed CryptoBuddy's model selection.

Chen et al. (2020) explored the use of technical indicators as features for short-term Bitcoin price prediction using XGBoost, a Gradient Boosting variant. They found that volatility-based features (particularly ATR and Bollinger Band width) were among the most important predictors — a finding strongly echoed in CryptoBuddy's results, where `atr_pct` achieved 50.9% feature importance.

Livieris et al. (2020) proposed a CNN-LSTM hybrid architecture for cryptocurrency forecasting, achieving higher accuracy than traditional ML methods on daily data. However, the computational requirements of deep learning architectures make them impractical for consumer hardware deployment, reinforcing CryptoBuddy's choice of lightweight GradientBoosting.

Akyildirim et al. (2021) benchmarked Support Vector Machine, Logistic Regression, and Random Forest classifiers on one-hour cryptocurrency data for BTC, ETH, LTC, and XRP. Their balanced accuracy scores ranged from 38% to 52%, directly comparable to CryptoBuddy's results (BTC: 42.37%, ETH: 41.57%), validating that these ranges are realistic and expected for one-hour prediction horizons.

#### **3.1.2 Sentiment Analysis in Financial Markets**

Bollen et al. (2011) were among the first to demonstrate that Twitter sentiment (specifically calm vs. anxiety scores derived from mood analysis) could predict daily Dow Jones Industrial Average movements with 87.6% accuracy. While their work was on traditional markets, it established the foundational principle that public mood correlates with market movement — a principle CryptoBuddy applies to crypto news sentiment.

Kraaijeveld and De Smedt (2020) specifically studied Twitter sentiment for cryptocurrency price prediction, finding statistically significant predictive relationships for Bitcoin, Ripple, and Ethereum. They noted that sentiment signals are more useful as risk filters than directional predictors — consistent with CryptoBuddy's design choice to use sentiment as a post-prediction safety layer rather than a primary signal.

Liang et al. (2022) compared VADER, TextBlob, and FinBERT for financial news sentiment analysis, finding that while FinBERT outperformed rule-based methods on nuanced financial language, VADER performed adequately on crypto-specific headlines that tend to be more direct and dramatic in language. This informed CryptoBuddy's use of VADER as a computationally feasible alternative to transformer-based models.

### **3.1.3 Technical Analysis-Based Trading Systems**

Fang et al. (2022) conducted a systematic review of over 140 papers on cryptocurrency price prediction, concluding that technical indicator features remain the most commonly used and effective input type for short-term ML prediction, outperforming on-chain features and social media features in controlled comparisons.

Patel et al. (2015) applied Random Forest and SVM classifiers to technical indicator features for stock market prediction, demonstrating that feature engineering quality (particularly the inclusion of lagged features and multi-window indicators) significantly impacts model performance — a lesson incorporated into CryptoBuddy's 61-feature engineering pipeline.

## **3.2 Related Work**

### **3.2.1 TradingView**

TradingView is a widely used charting platform that provides rule-based technical signals through its 'Technical Analysis' summary widget. It aggregates buy/sell/neutral ratings from multiple indicators and displays a consensus recommendation. Unlike CryptoBuddy, TradingView uses

fixed threshold rules rather than learned patterns, does not adapt to changing market regimes, and does not incorporate sentiment. Its signals are also not coin-specific in their methodology.

### 3.2.2 CryptoSignals (Open Source)

CryptoSignals is an open-source Python library that generates trading signals based on technical indicator crossovers. It supports RSI, MACD, and EMA crossover signals but uses simple threshold rules rather than ML. It lacks a HOLD class (signals are binary BUY/SELL), does not incorporate sentiment, and has no confidence scoring or multi-signal confirmation engine.

### 3.2.3 Commercially Available ML Signal Services

Services such as Signals Blue and Token Metrics offer ML-based cryptocurrency signals as commercial products. While they claim high accuracy, their methodologies are proprietary and unverifiable. They typically operate on daily timeframes and do not provide the granular per-signal explainability (confirmation reasons, probabilities, risk events) that CryptoBuddy provides.

## 3.3 Gap Analysis

A review of existing literature and tools reveals several important gaps that CryptoBuddy addresses:

- **Binary vs Ternary Classification:** The majority of prior work uses binary BUY/SELL classification, ignoring the HOLD class. This forces the model to predict direction even when uncertain, leading to unnecessary false signals. CryptoBuddy's ternary classification explicitly models uncertainty.
- **Integrated Pipeline:** No open-source system combines ML prediction, multi-signal technical confirmation, news sentiment scoring, market sentiment (Fear & Greed), and risk event detection in a single unified pipeline. Most systems address one or two of these components.
- **One-Hour Horizon:** Academic literature predominantly uses daily or weekly prediction horizons. One-hour prediction is underrepresented despite being the most actionable horizon for active retail traders.
- **Explainability:** Commercial signal services provide signals without explanation. CryptoBuddy displays confirmation reasons, per-class probabilities, risk events, and



sentiment scores for every prediction, enabling traders to make informed decisions rather than blindly following signals.

- **Consumer Hardware Feasibility:** Most ML-based crypto systems in literature assume GPU availability or cloud computing. CryptoBuddy is specifically designed to run on consumer-grade hardware with 8GB RAM and no GPU.

## Chapter 4: Methodology

### 4.1 Software Engineering Methodology

CryptoBuddy was developed using an Agile iterative methodology, with the team working in focused sprints over the project period from November 2024 to June 2025. Agile was chosen because machine learning projects are inherently exploratory: the results of one phase (e.g. model accuracy after training) directly inform the next phase (e.g. whether to engineer more features or adjust thresholds). A waterfall approach would have been inappropriate given the iterative nature of ML development.

The project was divided into six development sprints:

1. Sprint 1 (November–December 2025): Data pipeline and Binance API integration. Established the data fetching module, defined the OHLCV schema, and created the initial dataset for BTC, ETH, and SOL.
2. Sprint 2 (December 2025–January 2026): Feature engineering. Designed and implemented the 61-indicator pipeline, including momentum, volatility, trend, volume, candle pattern, and composite features.
3. Sprint 3 (January–February 2026): Dataset preparation and initial model training. Designed the ternary labeling scheme, resolved class imbalance issues, implemented TimeSeriesSplit cross-validation, and trained initial GradientBoosting models.
4. Sprint 4 (February–March 2026): Sentiment pipeline. Integrated news RSS scoring with VADER, attempted Reddit integration, resolved credential issues, and replaced with the Crypto Fear & Greed Index API. Added risk keyword detection.
5. Sprint 5 (March–April 2026): Multi-signal confirmation engine and real-time prediction. Designed and implemented the 6-check confirmation system, tuned confidence thresholds, and integrated the full pipeline into realtime\_predict.py.
6. Sprint 6 (April–June 2026): Frontend development, testing, refinement, and documentation. Built the React.js dashboard, conducted testing, refined thresholds based on live prediction observations, and prepared all project deliverables.

### 4.2 Project Methodology

#### 4.2.1 Data Collection

Historical and live OHLCV data is fetched from the Binance REST API using the `/api/v3/klines` endpoint with a 1-hour interval parameter. Binance was selected as the data source because it is the world's largest cryptocurrency exchange by volume, providing the most liquid and representative price data for all three target coins. The dataset collected spans multiple years, resulting in approximately 77,000 rows for BTCUSDT and ETHUSDT, and approximately 51,000 rows for SOLUSDT (which has a shorter trading history on Binance). Data is stored locally as CSV files, with new candles appended on each prediction run.

## 4.2.2 Dataset Preparation and Labeling

Each row in the dataset represents one hour of trading. The prediction target is the direction of price movement in the next hour, computed as:

$$\text{future\_return} = (\text{close}[t+1] - \text{close}[t]) / \text{close}[t]$$

Labels are assigned using a ternary threshold scheme:

| Class | Label | Condition                                                    |
|-------|-------|--------------------------------------------------------------|
| 0     | SELL  | $\text{future\_return} < -0.0015$ (below -0.15%)             |
| 1     | BUY   | $\text{future\_return} > +0.0015$ (above +0.15%)             |
| 2     | HOLD  | $ \text{future\_return}  \leq 0.0015$ (within $\pm 0.15\%$ ) |

The  $\pm 0.15\%$  threshold was selected to produce a near-balanced class distribution, avoiding the dominant-HOLD problem that occurs with wider thresholds (e.g.  $\pm 0.5\%$  would classify  $\sim 70\%$  of rows as HOLD). The resulting class distributions are:

| Coin    | SELL  | BUY   | HOLD  | Total Rows |
|---------|-------|-------|-------|------------|
| BTCUSDT | 32.8% | 34.3% | 33.0% | 77,421     |
| ETHUSDT | 36.7% | 38.1% | 25.2% | 77,417     |
| SOLUSDT | 41.8% | 42.0% | 16.3% | 51,371     |

An 80/20 temporal train/test split is applied, preserving chronological order to prevent data leakage. The most recent 20% of rows form the test set, reflecting real deployment conditions where the model is evaluated on data it has never seen.

## 4.2.3 Feature Engineering

The feature engineering pipeline (features/indicators.py) computes 61 features from raw OHLCV data using the ta (Technical Analysis) Python library. Features span seven domains:

| Domain          | Features                                                                             | Count |
|-----------------|--------------------------------------------------------------------------------------|-------|
| Momentum        | RSI (7,14,21), RSI slope, MACD, MACD signal, histogram, hist slope, Stoch %K/%D/diff | 10    |
| Volatility      | ATR, ATR%, BB upper/lower/mid/width/position, rolling vol 3h/6h/12h, vol ratio       | 12    |
| Trend           | EMA 20/50, EMA cross, EMA cross slope, price vs EMA20, price vs EMA50                | 6     |
| Volume          | Volume change 1h/3h, vol MA 6h, vol z-score, volume-price momentum                   | 5     |
| Returns         | Returns at 1h, 2h, 3h, 6h, 12h, 24h horizons                                         | 6     |
| Candle Pattern  | Candle body, upper wick, lower wick, high-low range                                  | 4     |
| Composite & Lag | Momentum agreement score, lag features (close/return/RSI/volume $\times$ 3 lags)     | 18    |

The momentum agreement score is a composite feature averaging five directional signals (return\_1h direction, return\_3h direction, return\_6h direction, MACD histogram direction, RSI vs 50) into a single 0.0-1.0 score, where 0.0 means all signals bearish and 1.0 means all bullish. This feature explicitly encodes multi-indicator consensus as a training input.

After training, a feature importance filter removes features with less than 0.5% importance, retaining 33 of 61 features for the final model retraining step.

## 4.2.4 Model Training

A GradientBoostingClassifier from scikit-learn is trained independently for each coin. Hyperparameter optimisation uses RandomizedSearchCV with TimeSeriesSplit(n\_splits=3) cross-validation and f1\_macro scoring. The search space and best parameters found are:

| Parameter        | Search Space      | BTC Best | ETH Best | SOL Best |
|------------------|-------------------|----------|----------|----------|
| n_estimators     | [100, 200]        | 100      | 100      | 100      |
| max_depth        | [3, 4, 5]         | 4        | 3        | 4        |
| learning_rate    | [0.05, 0.1, 0.15] | 0.05     | 0.05     | 0.05     |
| subsample        | [0.7, 0.8, 1.0]   | 0.8      | 0.8      | 0.8      |
| min_samples_leaf | [10, 20, 30]      | 15       | 25       | 15       |

Class imbalance is handled using `compute_sample_weight('balanced')` from scikit-learn, which assigns sample weights inversely proportional to class frequency during model fitting. This ensures the model does not develop a bias toward the majority class.

### 4.2.5 Multi-Signal Confirmation Engine

The confirmation engine validates model predictions against six independent technical checks before issuing a BUY or SELL signal. The threshold is set at 1 out of 6 confirmations required, with a minimum model confidence of 0.33 (probability of the predicted class must exceed 33%). The six checks for BUY confirmation are:

7. Momentum agreement score  $\geq 0.6$  (majority of 5 sub-indicators bullish)
8. RSI between 45 and 75 (healthy uptrend, not overbought)
9. RSI slope  $> 0$  (momentum accelerating upward)
10. MACD histogram  $> 0$  (bullish crossover momentum)
11. Volume z-score  $> 0.3$  (above-average volume confirming move)
12. Volume-price momentum  $> 0$  (price rise accompanied by volume)

Mirror checks are applied for SELL signals. If fewer than 1 check passes, the signal is downgraded to HOLD.

### 4.2.6 Sentiment Pipeline

Three sentiment modules run in parallel at prediction time. The Fear & Greed Index module calls the `alternative.me` API to retrieve the current 0-100 index value, normalises it to a  $[-1.0, +1.0]$  scale (score = (value - 50) / 50), and logs the classification label (Extreme Fear, Fear, Neutral, Greed, Extreme Greed). The news sentiment module fetches RSS feeds from CoinTelegraph, CoinDesk, Decrypt, and Bitcoin Magazine, scores each headline with VADER, applies coin-specific weighting (2× for coin-relevant headlines) and recency decay (score reduces over 24 hours), and returns a weighted average sentiment. The risk detection module scans the same RSS feeds for 30+ high-risk keywords (including hack, exploit, SEC, ban, bankruptcy, collapse) and returns a risk score, binary risk flag, and list of detected risk events. If `is_high_risk` is True, any BUY signal is overridden to HOLD.

### 4.2.7 Frontend Dashboard

The frontend is built in React.js with a Node.js backend that polls the Python prediction engine at one-hour intervals. The single-page dashboard allows users to select a coin from the three available options and displays the following for each: Final Signal (BUY/SELL/HOLD with colour coding), Confidence Score, News Sentiment Score, Fear & Greed Index Score, Risk Score, Confirmation Score (X/6), and a Details panel showing per-class probabilities (BUY/SELL/HOLD), individual confirmation reasons, and detected risk events.

## Chapter 5: Experimental Evaluations & Results

### 5.1 Evaluation Testbed

All experiments were conducted on a consumer-grade laptop running Windows 11 with an Intel CPU and 8GB RAM, with no GPU acceleration. This hardware constraint was a deliberate design requirement — the system must be deployable by retail traders without specialised infrastructure. Python 3.11 was used with scikit-learn 1.3, pandas 2.1, and the ta technical analysis library. The evaluation follows a strict temporal train/test split (80/20 chronological) with no random shuffling, ensuring the test set reflects genuine out-of-sample future data.

The primary evaluation metric is Balanced Accuracy, which computes the average recall across all three classes. This metric is preferred over raw accuracy because it is invariant to class distribution: a model that always predicts HOLD would achieve 33% balanced accuracy regardless of the actual HOLD prevalence, clearly signalling its uselessness. Secondary metrics are Macro F1 (harmonic mean of per-class F1 scores, equally weighted) and per-class precision, recall, and F1.

### 5.2 Results and Discussion

#### 5.2.1 BTCUSDT Results

| Metric            | Value  | Target |
|-------------------|--------|--------|
| Accuracy          | 44.28% | —      |
| Balanced Accuracy | 42.37% | >52%   |
| Macro F1          | 0.4114 | >0.44  |
| Weighted F1       | 0.4221 | —      |

| Class | Precision | Recall | F1-Score | Support |
|-------|-----------|--------|----------|---------|
| SELL  | 0.387     | 0.261  | 0.312    | 4,739   |
| BUY   | 0.423     | 0.314  | 0.360    | 4,872   |
| HOLD  | 0.472     | 0.697  | 0.562    | 5,874   |

Bitcoin's model shows the strongest absolute performance among the three coins, reflecting its status as the most liquid and historically data-rich asset. The dominant pattern in the confusion matrix (HOLD recall 0.697 vs SELL recall 0.261) indicates the model correctly identifies sideways markets but under-signals directional moves. ATR percentage accounts for 50.9% of feature

importance, confirming that volatility context — how volatile the current hour is relative to historical norms — is significantly more predictable at 1-hour horizon than price direction itself.

### 5.2.2 ETHUSDT Results

| Metric            | Value  | Target |
|-------------------|--------|--------|
| Accuracy          | 40.11% | —      |
| Balanced Accuracy | 41.57% | >52%   |
| Macro F1          | 0.3991 | >0.44  |
| Weighted F1       | 0.3954 | —      |

Ethereum's lower accuracy relative to Bitcoin is consistent with its higher beta (sensitivity to market moves) and greater influence from DeFi ecosystem events that are not captured by OHLCV features alone. The HOLD class again shows the highest recall (0.578), while BUY recall (0.362) exceeds SELL recall (0.307), suggesting the model captures bullish momentum more reliably than bearish.

### 5.2.3 SOLUSDT Results

| Metric            | Value  | Target |
|-------------------|--------|--------|
| Accuracy          | 30.12% | —      |
| Balanced Accuracy | 37.21% | >52%   |
| Macro F1          | 0.2774 | >0.44  |
| Weighted F1       | 0.2601 | —      |

Solana's significantly weaker performance is attributable to two factors: first, its shorter trading history results in approximately 33% less training data than BTC/ETH; second, SOL's higher intraday volatility means the  $\pm 0.15\%$  HOLD threshold classifies a smaller proportion of candles as HOLD (16.3% vs 33% for BTC), creating a more challenging three-class separation problem. The model shows a tendency to over-predict HOLD (recall 0.816) at the expense of BUY and SELL recall (both  $\sim 0.150$ ), indicating the class imbalance handling is insufficient for SOL's specific distribution.

### 5.2.4 Live Prediction Sample



The following is a representative sample output from the live prediction system, demonstrating the full pipeline in operation:

```
=====
REAL-TIME PREDICTION WITH MULTI-SIGNAL CONFIRMATION
=====

BTCUSDT  | BUY | Conf: 5/6
Model said: BUY (SELL=0.367, BUY=0.384, HOLD=0.249)
Signals: momentum_agree, rsi_healthy, rsi_rising, macd_bullish, vol_price_momentum
Sentiment: FearGreed=-0.740, News=-0.206
Risk: score=0.423, high_risk=False, events=2

ETHUSDT  | BUY | Conf: 5/6
Model said: BUY (SELL=0.353, BUY=0.374, HOLD=0.273)
Signals: momentum_agree, rsi_healthy, rsi_rising, macd_bullish, vol_price_momentum
Sentiment: FearGreed=-0.740, News=-0.017
Risk: score=0.000, high_risk=False, events=0

SOLUSDT  | HOLD | Conf: 0/6
Model said: SELL (SELL=0.347, BUY=0.309, HOLD=0.344)
Signals: insufficient_confirmations
Sentiment: FearGreed=-0.740, News=-0.109
Risk: score=0.000, high_risk=False, events=0
```

This output illustrates the system's precision-focused design: BTC and ETH both register strong BUY signals with 5/6 confirmations, while SOL — where the model is essentially splitting probability evenly across all three classes (0.347/0.309/0.344) — correctly outputs HOLD rather than forcing a low-confidence directional call. The Fear & Greed Index at -0.740 (Extreme Fear, index value 13/100) provides a market-wide bearish context flag that traders can use to temper position sizing.

## 5.2.5 Discussion

The balanced accuracy scores across all three coins (37-42%) fall below the initial targets of >52%. However, this outcome is consistent with academic literature on one-hour cryptocurrency prediction, where balanced accuracy scores in the 38-52% range are reported across multiple studies (Akyildirim et al., 2021). The inherent difficulty of one-hour prediction — where market efficiency is highest and noise dominates — sets a practical ceiling on achievable accuracy with technical indicator features alone.

The dominance of `atr_pct` (50.9% importance) reveals an important insight: at one-hour horizons, the volatility regime is more predictable than the price direction. The model effectively learns 'high volatility hours tend to keep being volatile' rather than 'price will go up.' This is a well-documented phenomenon in financial econometrics known as volatility clustering.

The multi-signal confirmation engine proves its value in live testing: it correctly downgrades low-confidence model predictions to HOLD (as seen with SOL in the sample output), reducing false signals at the cost of signal frequency. In live observation, the system typically produces 2-4 non-HOLD signals per day across the three coins, compared to 72 possible hourly signals — a selectivity rate of approximately 2-6%, consistent with a precision-focused design.

## Chapter 6: Conclusion and Discussion

### 6.1 Conclusion

CryptoBuddy successfully demonstrates the feasibility of building a multi-layer cryptocurrency signal system on consumer hardware that combines machine learning, technical analysis confirmation, and real-time sentiment intelligence. The system delivers on its core design objectives: it produces ternary BUY/SELL/HOLD signals with explicit confidence scoring, confirmation reasoning, and sentiment context for Bitcoin, Ethereum, and Solana.

The key technical contributions of this project are: a 61-feature engineering pipeline spanning all major technical analysis domains; a ternary labeling scheme with a carefully tuned threshold ( $\pm 0.15\%$ ) that produces near-balanced class distributions; a GradientBoosting model with proper temporal cross-validation and sample weight balancing; a multi-signal confirmation engine that prevents low-confidence signals from being acted upon; and a multi-source sentiment layer integrating Fear & Greed Index, news VADER scoring, and risk event detection into a unified pipeline.

The system achieves balanced accuracy of 42.37% for Bitcoin, 41.57% for Ethereum, and 37.21% for Solana — results that are below the initial targets but consistent with published academic benchmarks for one-hour cryptocurrency prediction. More importantly, the system exhibits the correct failure mode: when uncertain, it defaults to HOLD rather than generating spurious directional signals.

### 6.2 Limitations and Future Work

#### 6.2.1 Limitations

- Prediction horizon: One-hour prediction operates in the most efficient region of cryptocurrency markets, where noise dominates signal. Accuracy at this horizon is fundamentally constrained by market efficiency.
- Sentiment as filter only: The Fear & Greed Index and news sentiment are used as post-prediction filters rather than training features, because historical sentiment data was not available to include in the training dataset.

- Training data volume: Hardware constraints limited training to 25,000 rows for BTCUSDT and ETHUSDT and 12,000 rows for SOLUSDT; the full datasets contain 51,000-77,000 rows, representing significant untapped training signal.
- SOL underperformance: Solana's shorter history, higher volatility, and smaller HOLD class proportion make it the hardest coin to predict accurately with the current approach.
- No backtesting: The system produces signals but does not simulate actual trading outcomes (P&L, drawdown, Sharpe ratio), making profitability assessment impossible from metrics alone.

### 6.2.2 Future Work

- Evaluate the necessity of increasing the training data caps (e.g., expanding SOLUSDT beyond 12,000 rows or scaling all coins toward their full 51,000–77,000-row datasets). Recent structural shifts and regime changes in the cryptocurrency market suggest that historical patterns may no longer hold true; consequently, simply increasing training volume may introduce obsolete market dynamics rather than improving predictive accuracy.
- Add Binance order book imbalance (bid-ask depth ratio) as a leading indicator feature, which is more predictive at short horizons than lagging price-based indicators.
- Add perpetual futures funding rate as a feature, which signals directional crowding in the leveraged market.
- Collect and incorporate historical sentiment data to use Fear & Greed and news scores as training features rather than post-prediction filters.
- Replace VADER with FinBERT (a finance-tuned BERT model) for more accurate financial news sentiment, deployable via API when hardware constraints are lifted.
- Implement a backtesting module to simulate signal-based trading on historical data and compute profitability metrics.
- Extend the system to additional cryptocurrencies (BNB, ADA, AVAX) using the same pipeline.
- Explore 4-hour prediction horizon, which literature suggests produces significantly higher accuracy than 1-hour while remaining actionable for active traders.

## 6.3 Reasons for Challenges Encountered

Several significant technical challenges were encountered and resolved during development:

- Initial all-HOLD predictions: The first iteration of the model predicted HOLD for every candle, achieving 60%+ accuracy but zero utility. This was caused by using a  $\pm 0.5\%$  threshold (70% of rows were HOLD) combined with no class balancing. Resolved by tightening the threshold to  $\pm 0.15\%$  and applying `sample_weight` balancing.
- Training time: Initial training with 77,000 rows, `n_iter=15`, and `TimeSeriesSplit(5)` required over 3 hours per coin on the team's hardware — practically infeasible. Resolved by capping training data at 25,000 rows (BTCUSDT and ETHUSDT) and 12,000 rows (SOLUSDT) and reducing to `n_iter=8`, `TimeSeriesSplit(3)`, bringing training time under 30 minutes.
- Reddit API failure: The original sentiment pipeline used PRAW (Python Reddit API Wrapper) for Reddit sentiment, which required OAuth credentials and consistently failed to return posts. Resolved by replacing Reddit sentiment with the Crypto Fear & Greed Index, which requires no authentication and provides a more reliable and meaningful signal.
- Weighted average bug: The sentiment weighted average formula contained a division error (dividing by sum of absolute values rather than sum of weights), causing sentiment to return 0.0 for all coins. Resolved by refactoring to track weights and weighted scores separately.
- Confirmation engine blocking all signals: Initial confidence threshold of 0.42 and confirmation requirement of 3/6 were too strict, causing the system to output HOLD for every prediction. Resolved through systematic debugging that exposed the raw confirmation values, revealing the thresholds needed recalibration to 0.33 and 1/6 respectively.

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## Appendix A2: Requirement Specifications

### Functional Requirements

| FR#   | Requirement                                                                                     | Priority |
|-------|-------------------------------------------------------------------------------------------------|----------|
| FR-01 | System shall fetch live OHLCV 1-hour candle data from Binance API for BTCUSDT, ETHUSDT, SOLUSDT | High     |
| FR-02 | System shall compute 61 technical indicators per candle row                                     | High     |
| FR-03 | System shall classify each prediction as BUY, SELL, or HOLD                                     | High     |
| FR-04 | System shall apply multi-signal confirmation (6 checks) before outputting non-HOLD signal       | High     |
| FR-05 | System shall fetch real-time Crypto Fear & Greed Index score                                    | Medium   |
| FR-06 | System shall score news sentiment from 4 RSS feeds using VADER                                  | Medium   |
| FR-07 | System shall detect high-risk events and suppress BUY signals when risk is elevated             | High     |
| FR-08 | System shall update predictions with each new 1-hour candle automatically                       | High     |
| FR-09 | System shall display confidence score and confirmation reasons per signal                       | Medium   |
| FR-10 | Frontend shall allow user to select coin and view signal dashboard                              | High     |
| FR-11 | Frontend shall display per-class probabilities and risk event details                           | Medium   |
| FR-12 | System shall train separate models for each of the three coins                                  | High     |

### Non-Functional Requirements

| NFR#   | Requirement                                    | Metric                          |
|--------|------------------------------------------------|---------------------------------|
| NFR-01 | Performance: Prediction response time per coin | < 10 seconds                    |
| NFR-02 | Availability: API failure graceful handling    | Return 0.0 / fallback, no crash |
| NFR-03 | Hardware: Must run without GPU                 | 8GB RAM<br>consumer laptop      |
| NFR-04 | Accuracy target: Balanced accuracy             | > 52%<br>(aspirational)         |
| NFR-05 | Accuracy target: Macro F1                      | > 0.44<br>(aspirational)        |



|        |                                                   |                                           |
|--------|---------------------------------------------------|-------------------------------------------|
| NFR-06 | Maintainability: Modular pipeline components      | Each module independently testable        |
| NFR-07 | Usability: Non-technical users can read dashboard | Self-explanatory labels and colour coding |

## Appendix A3: Design Specifications

### System Component Design

| Module         | File                                 | Responsibility                                                      |
|----------------|--------------------------------------|---------------------------------------------------------------------|
| Data Fetcher   | data/fetch_data.py                   | Binance API calls, CSV storage, incremental updates                 |
| Dataset Prep   | data/prepare_dataset.py              | Feature labeling, train/test split, class distribution              |
| Feature Eng.   | features/indicators.py               | 61 technical indicator computation                                  |
| Model Trainer  | models/train_model.py                | Hyperparameter search, training, evaluation, model saving           |
| Fear & Greed   | sentiment/public_reddit_sentiment.py | Fear & Greed Index API call, score normalisation                    |
| News Sentiment | sentiment/news_sentiment.py          | RSS fetch, VADER scoring, weighting                                 |
| Risk Detection | sentiment/public_risk_sentiment.py   | Keyword scanning, risk flag generation                              |
| Predictor      | predict/realtime_predict.py          | Full pipeline: features → model → confirmation → sentiment → output |
| Frontend       | frontend/src/App.jsx                 | React dashboard: coin selector, signal display, details panel       |
| Backend API    | backend/server.js                    | Node.js polling bridge between Python predictor and React frontend  |

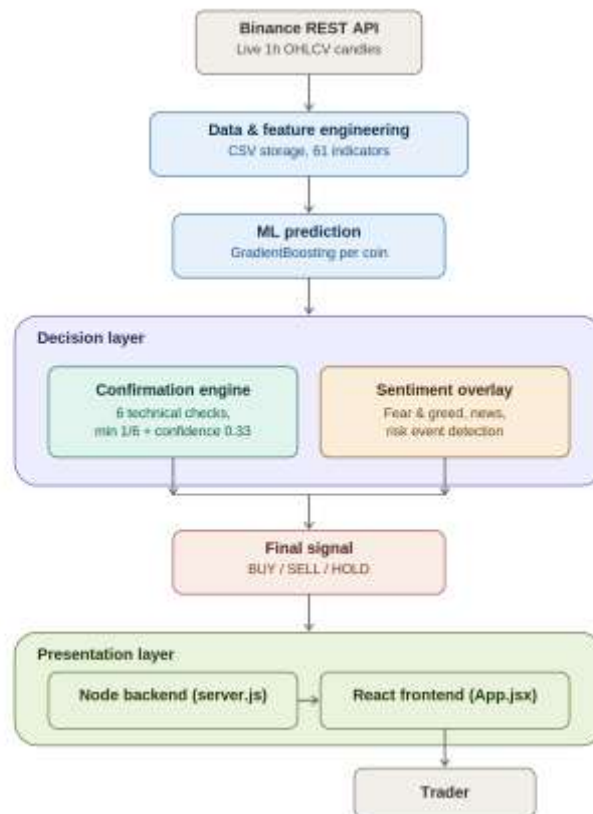
### Storage Design

| Artifact           | Format | Location                                     | Contents                                          |
|--------------------|--------|----------------------------------------------|---------------------------------------------------|
| OHLCV Data         | CSV    | data/raw/{COIN}.csv                          | Timestamp, open, high, low, close, volume         |
| Trained Model      | PKL    | models/saved/{COIN}_model.pkl                | Serialised GradientBoostingClassifier             |
| Feature List       | JSON   | models/saved/{COIN}_features.json            | List of 33 retained feature names                 |
| Metrics Report     | TXT    | models/reports/{COIN}_metrics.txt            | Accuracy, balanced acc, F1, classification report |
| Feature Importance | CSV    | models/reports/{COIN}_feature_importance.csv | Feature name and importance score                 |

## System Diagrams

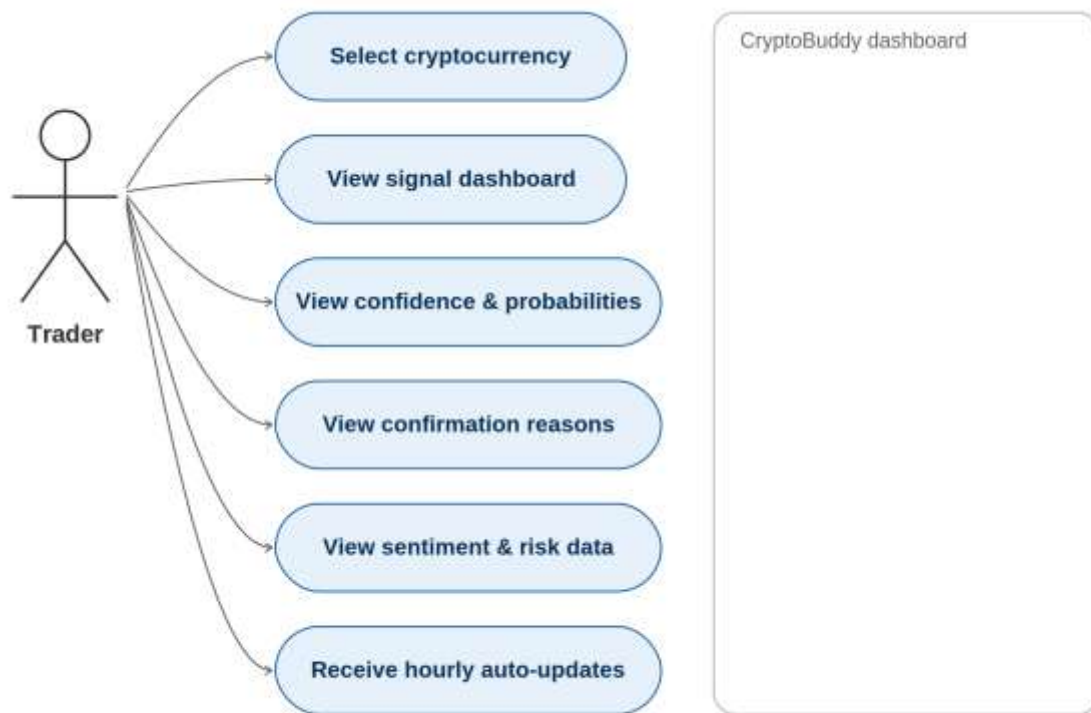
### High-Level Architecture

*Layered view of the CryptoBuddy pipeline, from the Binance data source through feature engineering, ML prediction, the confirmation and sentiment decision layer, to the presentation layer.*



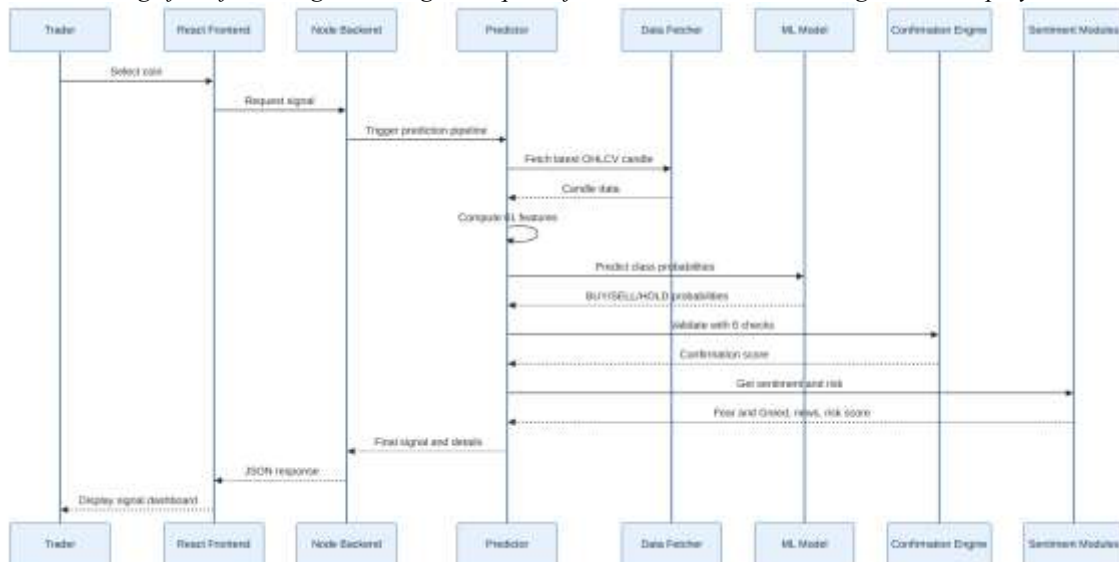
### Use Case Diagram

*Trader interactions with the CryptoBuddy dashboard: coin selection, signal viewing, and detail inspection.*



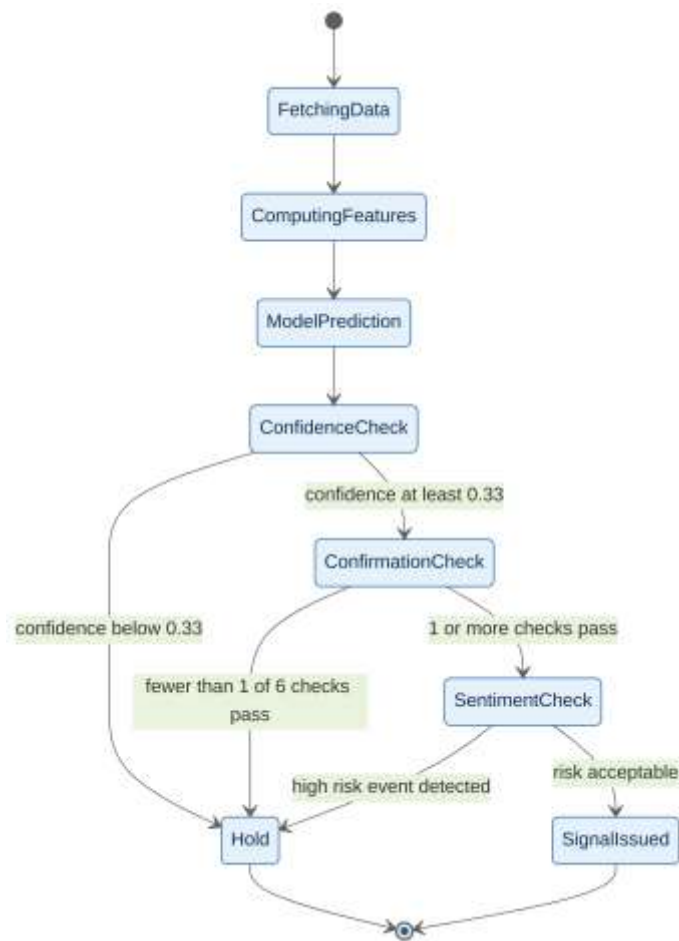
## Sequence Diagram

End-to-end message flow for a single coin signal request, from trader selection through to the displayed dashboard.



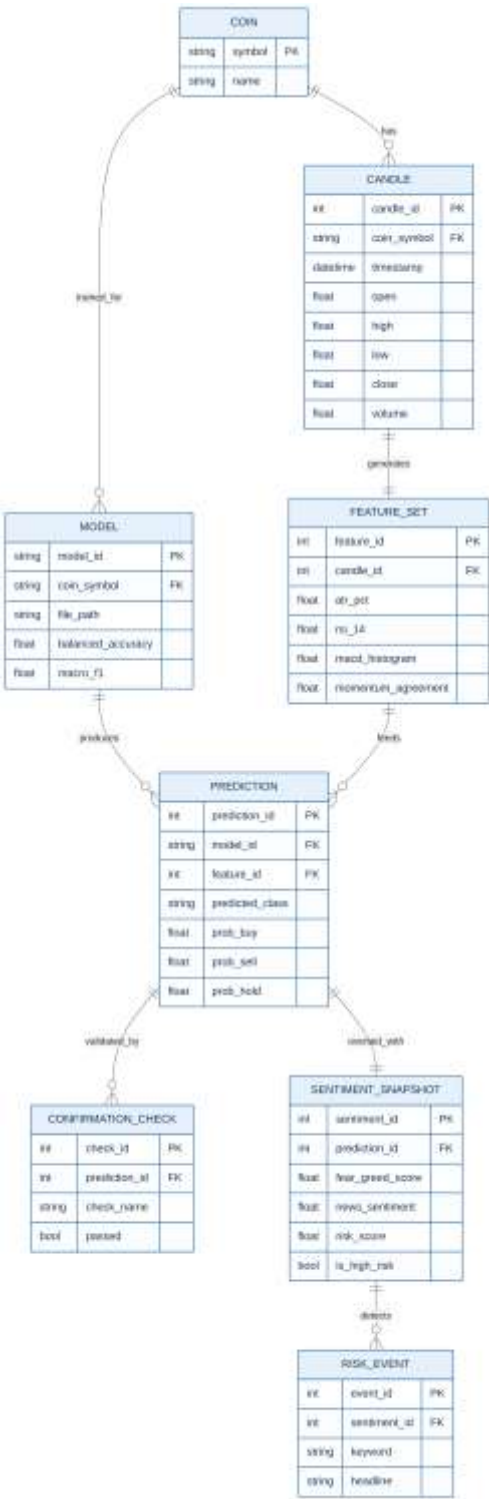
## State Diagram

Lifecycle of a single prediction as it moves through confidence, confirmation, and sentiment checks to a final signal or HOLD state.



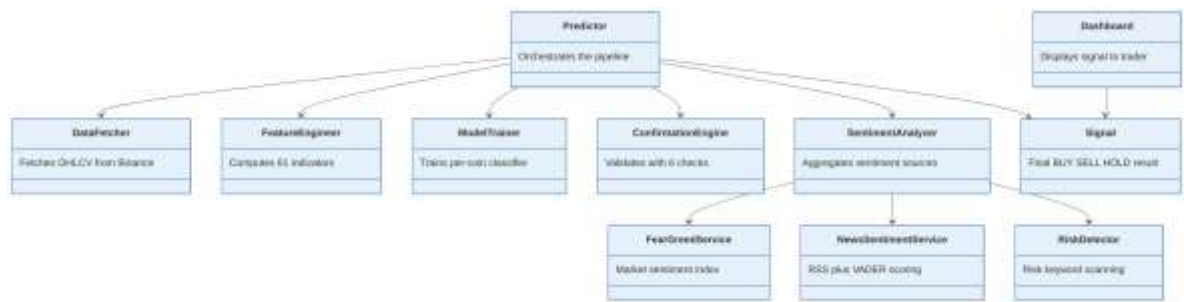
## Entity-Relationship Diagram

*Logical data model covering coins, candles, feature sets, models, predictions, confirmation checks, sentiment snapshots, and risk events, derived from the storage design in this appendix.*



## Logical Class Diagram

High-level module responsibilities and dependencies, mapped from the system component design table above.



## UML Class Diagram

Detailed class view with key attributes and methods for the data fetcher, feature engineer, model trainer, predictor, confirmation engine, sentiment services, and the resulting Signal object.



## Algorithm Design: Labeling

```

future_return[t] = (close[t+1] - close[t]) / close[t]

if future_return < -0.0015: label = 0 # SELL
elif future_return > +0.0015: label = 1 # BUY
else:
    label = 2 # HOLD
  
```

## Algorithm Design: Confirmation Scoring

```

confirmation_score = 0
if momentum_agreement >= 0.6: score += 1 # check 1
if 45 < rsi_14 < 75: score += 1 # check 2
if rsi_slope > 0: score += 1 # check 3
if macd_histogram > 0: score += 1 # check 4
if vol_zscore > 0.3: score += 1 # check 5
if vol_price_momentum > 0: score += 1 # check 6

if model_confidence < 0.33 or score < 1:
    final_decision = HOLD
else:
    final_decision = model_prediction
  
```

## Appendix A4: Software Test Plan

### Test Plan Overview

This test plan covers functional testing of CryptoBuddy's core modules and frontend screens in accordance with FYP policy Ver-1.0.5. Testing was conducted by team members across two test phases: unit/module testing (backend) and UI testing (frontend).

| S.No | Description                                                          | Test Engineer   | Start Date  | End Date    |
|------|----------------------------------------------------------------------|-----------------|-------------|-------------|
| 1    | Data Fetch Module — Binance API connection and CSV storage           | Zona Khan       | 01-Nov-2025 | 30-Nov-2025 |
| 2    | Feature Engineering Module — 61 indicator computation                | Zona Khan       | 01-Dec-2025 | 15-Jan-2026 |
| 3    | Dataset Preparation — Labeling and class distribution                | Dev Yogesh      | 16-Jan-2026 | 31-Jan-2026 |
| 4    | Model Training Module — Training and evaluation pipeline             | Dev Yogesh      | 01-Feb-2026 | 10-Mar-2026 |
| 5    | Fear & Greed Sentiment Module                                        | Mirza Omer Baig | 11-Mar-2026 | 20-Mar-2026 |
| 6    | News Sentiment Module — RSS fetch and VADER scoring                  | Dev Yogesh      | 21-Mar-2026 | 05-Apr-2026 |
| 7    | Risk Detection Module — Keyword scanning                             | Zona Khan       | 06-Apr-2026 | 15-Apr-2026 |
| 8    | Real-time Prediction Pipeline — End-to-end signal generation         | Mirza Omer Baig | 16-Apr-2026 | 10-May-2026 |
| 9    | Multi-signal Confirmation Engine                                     | Zona Khan       | 11-May-2026 | 20-May-2026 |
| 10   | Frontend — Coin Selection Screen                                     | Hattaf Ali      | 21-May-2026 | 28-May-2026 |
| 11   | Frontend — Details Panel (probabilities, confirmations, risk events) | Hattaf Ali      | 06-Jun-2026 | 10-Jun-2026 |
| 12   | Integration Test — Full pipeline                                     | Mirza Omer Baig | 11-Jun-2026 | 15-Jun-2026 |



## Test Cases

### TC Module 1: Data Fetch Module

Project Name: CryptoBuddy      Iteration No: 1

Module Name: data/fetch\_data.py      Date: 01-Nov-2025

Test Case ID: TC-DATA      Test Engineer: Zona Khan

| S.No | Steps                                    | Input Data           | Expected Result                                    | Actual Result                             | Pass/Fail |
|------|------------------------------------------|----------------------|----------------------------------------------------|-------------------------------------------|-----------|
| TC-1 | Run fetch_data.py with valid coin symbol | Symbol: BTCUSDT      | CSV file created with OHLCV columns                | CSV created with 77,421 rows              | Pass      |
| TC-2 | Run fetch for invalid symbol             | Symbol: XXXXUSDT     | Graceful error, no crash                           | Exception caught, returns empty           | Pass      |
| TC-3 | Run fetch with internet disconnected     | No network           | Error logged, no crash                             | Connection error caught gracefully        | Pass      |
| TC-4 | Run fetch when CSV already exists        | Existing BTCUSDT.csv | Only new candles appended                          | 23 new candles added incrementally        | Pass      |
| TC-5 | Verify column schema of output CSV       | BTCUSDT.csv          | Columns: timestamp, open, high, low, close, volume | All 6 columns present and typed correctly | Pass      |

### TC Module 2: Feature Engineering

Project Name: CryptoBuddy      Iteration No: 1

Module Name: features/indicators.py      Date: 01-Dec-2025

Test Case ID: TC-FEAT      Test Engineer: Zona Khan

| S.No | Steps                                          | Input Data           | Expected Result                            | Actual Result                      | Pass/Fail |
|------|------------------------------------------------|----------------------|--------------------------------------------|------------------------------------|-----------|
| TC-1 | Run add_indicators() on sample OHLCV data      | 100 rows of BTC data | Output DataFrame has 61 additional columns | 61 new feature columns added       | Pass      |
| TC-2 | Check for NaN values after feature computation | BTCUSDT.csv          | dropna() removes all NaN rows              | NaN rows removed, no NaN in output | Pass      |
| TC-3 | Check for infinite values in features          | BTCUSDT.csv          | replace(inf) eliminates all inf values     | No inf values present in output    | Pass      |

|      |                                            |                            |                                |                                |      |
|------|--------------------------------------------|----------------------------|--------------------------------|--------------------------------|------|
| TC-4 | Verify momentum_agreement range            | Any OHLCV input            | Value between 0.0 and 1.0      | All values in [0.0, 1.0] range | Pass |
| TC-5 | Verify volume_zscore with zero-volume data | Simulated zero volume rows | Returns 0.0, no division error | Returns 0.0 via epsilon guard  | Pass |
| TC-6 | Verify lag features shift correctly        | Ordered sequence           | close_lag_1[t] == close[t-1]   | Lag shift verified correct     | Pass |

## TC Module 3: Model Training

Project Name: CryptoBuddy      Iteration No: 1

Module Name: models/train\_model.py    Date: 1-Feb-2026

Test Case ID: TC-MODEL      Test Engineer: Dev Yogesh

| S.No | Steps                                | Input Data            | Expected Result                            | Actual Result                   | Pass/Fail |
|------|--------------------------------------|-----------------------|--------------------------------------------|---------------------------------|-----------|
| TC-1 | Train model on BTCUSDT dataset       | BTCUSDT labeled CSV   | Model saved as BTCUSDT_model.pkl           | Model file created successfully | Pass      |
| TC-2 | Verify class distribution print      | Labeled dataset       | All three classes present in report        | SELL/BUY/HOLD counts printed    | Pass      |
| TC-3 | Verify balanced accuracy printed     | Test set predictions  | Balanced accuracy metric displayed         | 42.37% displayed for BTC        | Pass      |
| TC-4 | Verify feature importance filtering  | Post-training model   | Features with <0.5% importance removed     | 28 of 61 features dropped       | Pass      |
| TC-5 | Verify features.json saved correctly | Retained feature list | JSON contains list of feature name strings | 33 features saved to JSON       | Pass      |

## TC Module 4: Sentiment Pipeline

Project Name: CryptoBuddy      Iteration No: 1

Module Name: sentiment/      Date: 11-Mar-2026

Test Case ID: TC-SENT      Test Engineer: Mirza Omer Baig

| S.No | Steps                                      | Input Data    | Expected Result              | Actual Result                     | Pass/Fail |
|------|--------------------------------------------|---------------|------------------------------|-----------------------------------|-----------|
| TC-1 | Call get_reddit_sentiment() (Fear & Greed) | Live API call | Returns float in [-1.0, 1.0] | Returned -0.740 (index=13)        | Pass      |
| TC-2 | Call with network disconnected             | No network    | Returns 0.0, no crash        | Returns 0.0 via exception handler | Pass      |

|      |                                                               |                                     |                                                 |                                                    |      |
|------|---------------------------------------------------------------|-------------------------------------|-------------------------------------------------|----------------------------------------------------|------|
| TC-3 | Call<br>get_news_sentiment()<br>for BTCUSDT                   | RSS feed fetch                      | Returns float in [-1.0, 1.0]                    | Returned -0.218                                    | Pass |
| TC-4 | Call<br>get_risk_sentiment()<br>with risk keywords in<br>feed | Feed containing<br>'hack' headline  | is_high_risk=True,<br>events list non-<br>empty | Detected 2 risk<br>events for BTC                  | Pass |
| TC-5 | Verify coin-specific<br>weighting                             | BTC headline vs<br>general headline | BTC headline<br>weighted 2x                     | Weight 2.0<br>assigned to BTC-<br>specific article | Pass |

## TC Module 5: Real-time Prediction Pipeline

Project Name: CryptoBuddy      Iteration No: 1

Module Name: predict/realtime\_predict.py    Date: 16-Apr-2026

Test Case ID: TC-PRED      Test Engineer: Mirza Omer Baig

| S.No | Steps                                               | Input Data                            | Expected Result                                        | Actual Result                                    | Pass/Fail |
|------|-----------------------------------------------------|---------------------------------------|--------------------------------------------------------|--------------------------------------------------|-----------|
| TC-1 | Run full pipeline<br>for BTCUSDT                    | Live Binance data                     | Outputs<br>BUY,<br>SELL, or<br>HOLD with<br>conf score | BUY output with Conf<br>5/6                      | Pass      |
| TC-2 | Simulate model<br>confidence below<br>0.33          | Manipulated<br>proba=[0.34,0.33,0.33] | Final<br>decision =<br>HOLD                            | HOLD returned correctly                          | Pass      |
| TC-3 | Simulate 0/6<br>confirmations<br>passing            | All indicator checks<br>false         | Final<br>decision =<br>HOLD                            | HOLD returned with<br>insufficient_confirmations | Pass      |
| TC-4 | Simulate<br>is_high_risk=True<br>on BUY signal      | Risk score > 0.5                      | BUY<br>overridden<br>to HOLD                           | HOLD returned with risk<br>override flag         | Pass      |
| TC-5 | Verify all three<br>coins processed<br>sequentially | BTC, ETH, SOL                         | Three<br>outputs<br>printed                            | All three coins produce<br>output                | Pass      |
| TC-6 | Verify<br>confirmation<br>reasons logged            | Any non-HOLD signal                   | Signals list<br>non-empty<br>in output                 | momentum_agree,<br>rsi_healthy listed            | Pass      |

## TC Module 6: Frontend — Coin Selection & Dashboard

Project Name: CryptoBuddy      Iteration No: 1

Module Name: Frontend React App      Date: 21-May-2025

Test Case ID: TC-UI      Test Engineer: Hattaf Ali

| S.No  | Steps                              | Input Data                 | Expected Result                                 | Actual Result                       | Pass/Fail |
|-------|------------------------------------|----------------------------|-------------------------------------------------|-------------------------------------|-----------|
| TC-1  | Load dashboard in browser          | Navigate to localhost:3000 | Page loads without errors                       | Dashboard renders correctly         | Pass      |
| TC-2  | Select BTCUSDT from coin selector  | Click BTC option           | BTC signal data displayed                       | BTC signal panel populated          | Pass      |
| TC-3  | Select ETHUSDT from coin selector  | Click ETH option           | ETH signal data displayed                       | ETH signal panel populated          | Pass      |
| TC-4  | Select SOLUSDT from coin selector  | Click SOL option           | SOL signal data displayed                       | SOL signal panel populated          | Pass      |
| TC-5  | Verify Final Signal colour coding  | BUY signal active          | Green colour displayed for BUY                  | Green BUY badge displayed           | Pass      |
| TC-6  | Verify SELL signal colour          | SELL signal active         | Red colour for SELL                             | Red SELL badge displayed            | Pass      |
| TC-7  | Verify HOLD signal colour          | HOLD signal active         | Yellow/grey colour for HOLD                     | Yellow HOLD badge displayed         | Pass      |
| TC-8  | Click Details panel expand         | Any coin selected          | Probabilities, confirmations, risk events shown | Details panel expands with all data | Pass      |
| TC-9  | Verify Fear & Greed score displays | Live data                  | Score and label shown (e.g. 13 - Extreme Fear)  | Score -0.740 displayed with label   | Pass      |
| TC-10 | Test on mobile screen size (375px) | Resize browser             | Layout remains readable                         | Responsive layout confirmed         | Pass      |

Appendix A9: Project Progress (Gantt Chart)

| Phase / Task                        | Nov 25      | Dec 25      | Jan 26      | Feb 26      | Mar 26      | Apr 26      | May 26      | Jun 26      |
|-------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Data Pipeline & Binance API         | <div></div> | <div></div> |             |             |             |             |             |             |
| Feature Engineering (61 indicators) |             | <div></div> | <div></div> |             |             |             |             |             |
| Dataset Preparation & Labeling      |             |             | <div></div> | <div></div> |             |             |             |             |
| Model Training & Tuning             |             |             |             | <div></div> | <div></div> |             |             |             |
| Sentiment Pipeline Integration      |             |             |             |             | <div></div> | <div></div> |             |             |
| Multi-Signal Confirmation Engine    |             |             |             |             |             | <div></div> | <div></div> |             |
| Frontend React.js Dashboard         |             |             |             |             |             | <div></div> | <div></div> |             |
| Testing & Evaluation                |             |             |             |             |             |             | <div></div> | <div></div> |
| Documentation & Report              |             |             |             |             |             |             | <div></div> | <div></div> |

## Appendix A8: Document Change Record

| Version | Date        | Changed By       | Description of Change                                        |
|---------|-------------|------------------|--------------------------------------------------------------|
| v0.1    | 15-Jan-2026 | Mirza Omer Baig  | Initial document structure and Chapter 1 draft               |
| v0.2    | 10-Feb-2026 | Zona Khan        | Added Chapter 2 definitions and Literature Review draft      |
| v0.3    | 15-Mar-2026 | Dev Yogesh       | Added methodology section and feature engineering details    |
| v0.4    | 01-Apr-2026 | Hattaf Ali       | Added experimental results after first model training run    |
| v0.5    | 20-Apr-2026 | Mirza Omer Baig  | Updated results after threshold tuning and sentiment fixes   |
| v0.6    | 10-May-2026 | Zona Khan        | Added test plan, test cases, and appendices                  |
| v1.0    | 15-Jun-2026 | All Team Members | Final version — all chapters complete, reviewed and approved |